Deep contextualized word representations

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Presenter: 성지석

1. Introduction

(1) Existing models like *Word2Vec* or *Glove* provides a single context-independent representations.

(2) Some translation based models (*CoVe*) which uses transfer learning to supervised tasks are limited by the size of parallel corpora.

(3) Rest of models like *TagLM*,utilizes representations from final layer of the model which essentially leave out information from the lower layers.

(4) we learn a linear combination of the vectors stacked above each input word for each end task, which markedly improves performance over just using the top LSTM layer.

(5) **higher-level LSTM** states capture context-dependent aspects of **word meaning** (e.g., they can be used without modification to perform well on supervised word sense disambiguation tasks) while **lower-level** states model aspects of **syntax** (e.g., they can be used to do part-of-speech tagging)

2. Recall: Accuracy vs F_1 score

- Accuracy vs F_1 score

	True	False
True	True Positive (TP)	False Negative (FN)
False	False Positive (FP)	True Negative (TN)

  - Accuracy: datasets가 symmetric 하여 FP와 FN이 거의 같은 경우, 사용하면 좋음.
    (높은 accuracy 를 보이면 해당 model 이 좋은 것)
    Accuracy = (TP+TN)/(TP+FP+FN+TN)

  - F_1 score: Uneven class distribution 을 지닐 때 좋음. 즉, FP 와 FN 의 cost 가 상당히 다른 경우.
    F_1 score = 2*(Recall*Precision)/(Recall+Precision)

source: https://blog.exsilio.com/all/accuracy-precision-recall-f1-score-interpretation-of-performance-measures/

2. Recall: Recall and Precision

- Recall and Precision

	True	False
True	True Positive (TP)	False Negative (FN)
False	False Positive (FP)	True Negative (TN)

  - Recall (Sensitivity): The ratio of correctly predicted positive observations to the all observations in actual classes.
    Recall = TP/(TP+FN)
  - Precision: The ratio of correctly predicted positive observations to the total predicted positive observations
    Precision = TP/(TP+FP)

source: https://blog.exsilio.com/all/accuracy-precision-recall-f1-score-interpretation-of-performance-measures/

2. Recall : Language Model

Dan Jurafsky

Probabilistic Language Modeling

- Goal: compute the probability of a sentence or sequence of words:

  $P(W) = P(w_1,w_2,w_3,w_4,w_5 \ldots w_n)$

- Related task: probability of an upcoming word:

  $P(w_5|w_1,w_2,w_3,w_4)$

- A model that computes either of these:

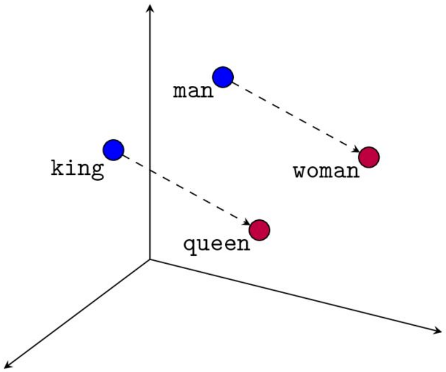
  $P(W)$ or $P(w_n|w_1,w_2 \ldots w_{n-1})$ is called a **language model**.

- Better: **the grammar** But **language model** or **LM** is standard

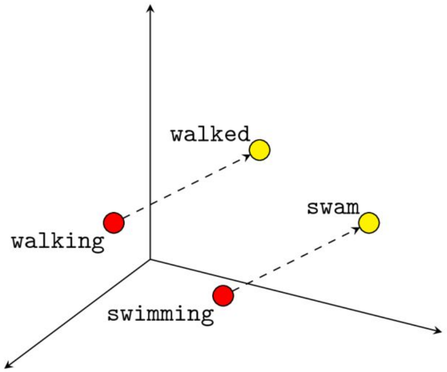
source: https://web.stanford.edu/~jurafsky/slp3/slides/LM_4.pdf

2 Recall : Word2Vec

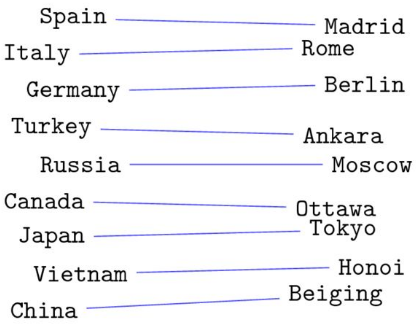
- Word를 vector로 변환하는 것
  - 단어들이 실수 공간에 흩어져 있음
- 수치적으로 해석 가능
  - 단어들 사이의 유사도를 측정
  - 단어들의 의미를 수식으로 계산 가능 (KING - MAN + WOMAN = QUEEN)



성별



시제



국가-수도

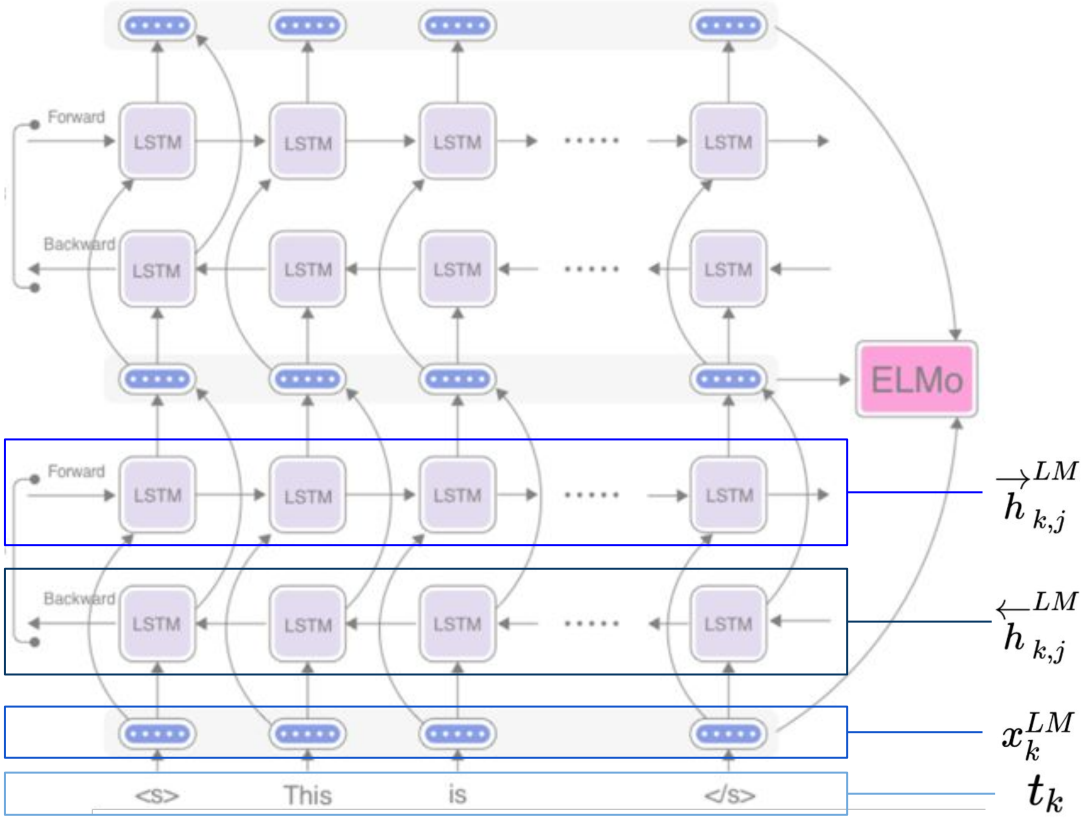
source: http://bit.ly/2XF4GSx / http://bit.ly/2S7GyCh

3 ELMo

- Structure

Layer 1

Layer 2



t_k: token
k: k-th token
j: j-th biLM layer

source: https://ireneli.eu/2018/12/17/elmo-in-practice/

3 ELMo

- Bidirectional language models

$$p(t_1, t_2, \ldots, t_N) = \prod_{k=1}^{N} p(t_k \mid t_1, t_2, \ldots, t_{k-1}).$$

$$p(t_1, t_2, \ldots, t_N) = \prod_{k=1}^{N} p(t_k \mid t_{k+1}, t_{k+2}, \ldots, t_N).$$

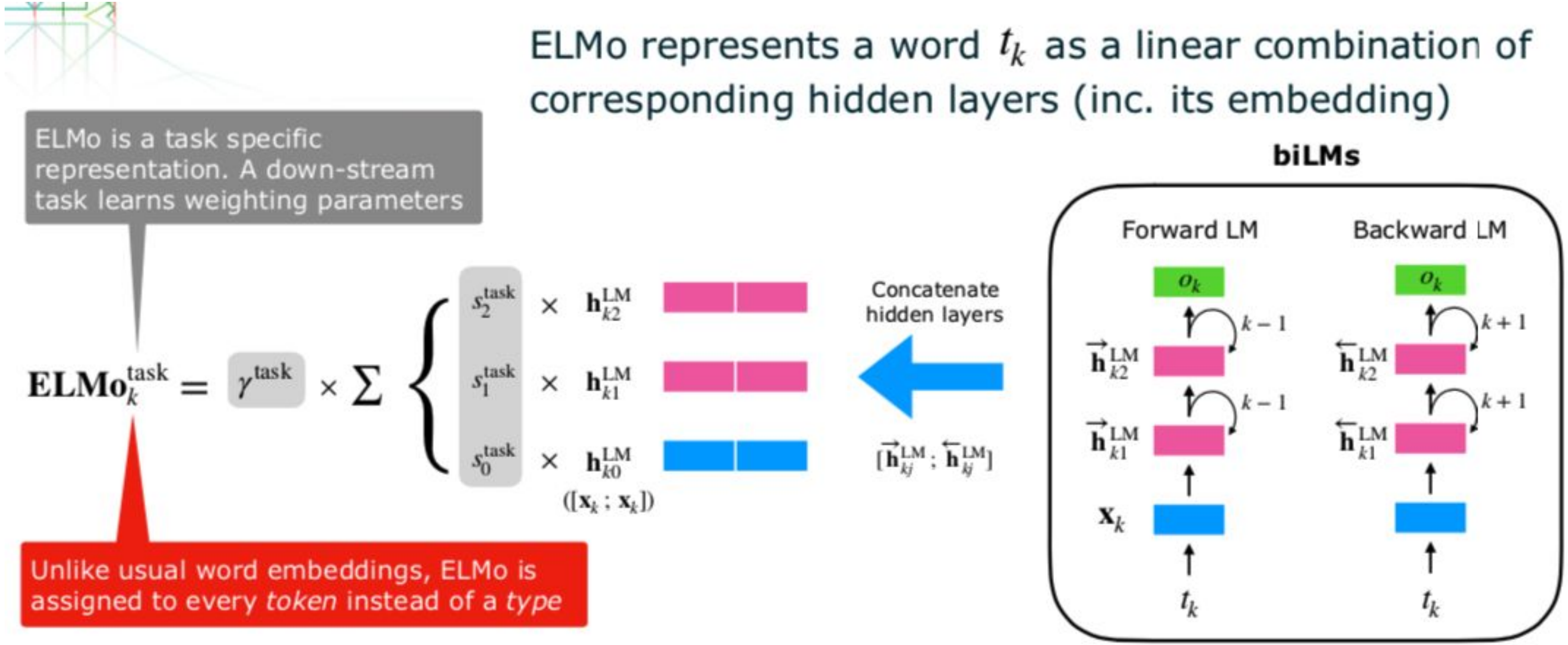
- 두 방향으로부터 나오는 확률의 합을 최대화 하도록 학습시킴

$$\sum_{k=1}^{N} \Big(\log p(t_k \mid t_1, \ldots, t_{k-1}; \Theta_x, \overrightarrow{\Theta}_{LSTM}, \Theta_s) + \log p(t_k \mid t_{k+1}, \ldots, t_N; \Theta_x, \overleftarrow{\Theta}_{LSTM}, \Theta_s) \Big).$$

* 인풋 벡터, 소프트맥스 레이어의 파라미터는 공유

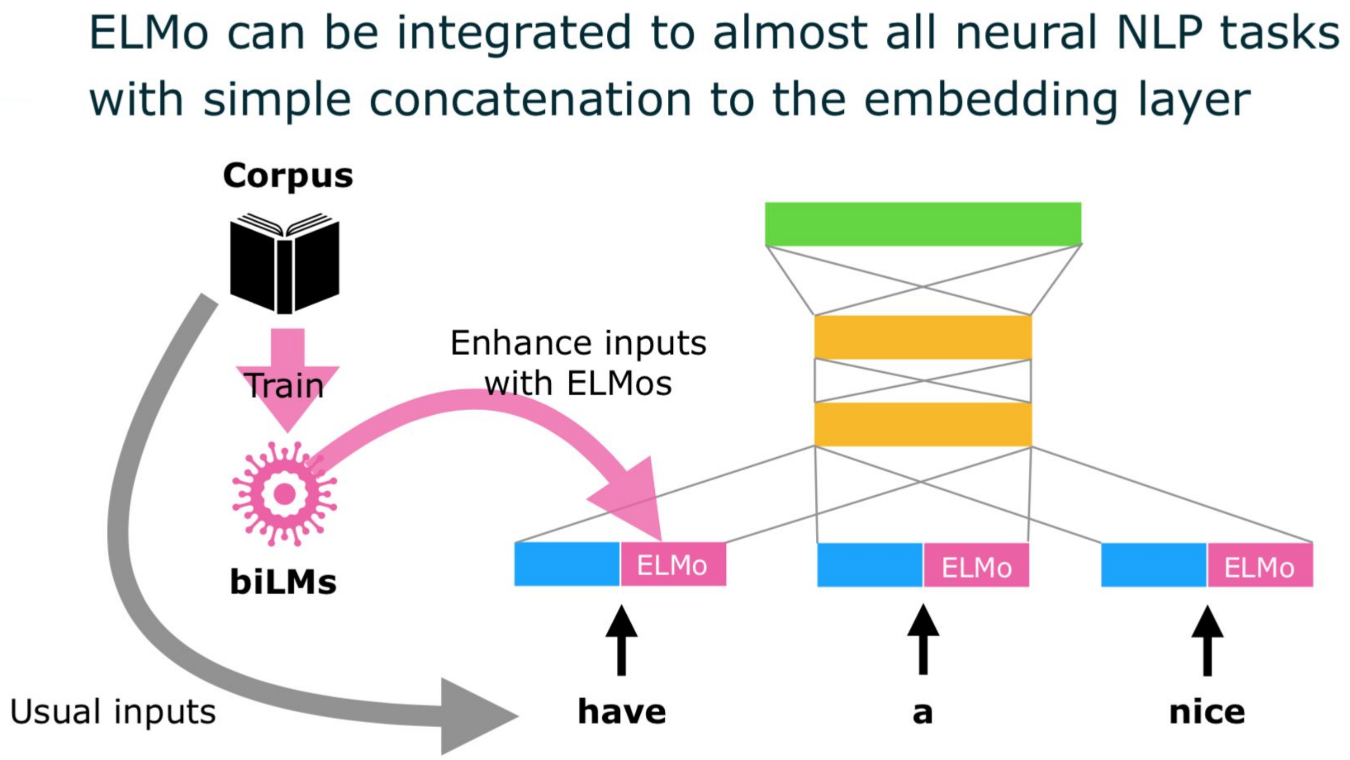
source : http://bit.ly/2FRy1yg

3 ELMo



source: https://www.slideshare.net/shuntaroy/a-review-of-deep-contextualized-word-representations-peters-2018

3 ELMo



source: https://www.slideshare.net/shuntaroy/a-review-of-deep-contextualized-word-representations-peters-2018

4 Evaluation

Task	Previous SOTA		Our baseline		ELMo + baseline	Increase (absolute/relative)
SQuAD	Liu et al. (2017)	84.4	81.1	⟶	85.8	4.7 / 24.9%
SNLI	Chen et al. (2017)	88.6	88.0	⟶	88.7 ± 0.17	0.7 / 5.8%
SRL	He et al. (2017)	81.7	81.4	⟶	84.6	3.2 / 17.2%
Coref	Lee et al. (2017)	67.2	67.2	⟶	70.4	3.2 / 9.8%
NER	Peters et al. (2017)	91.93 ± 0.19	90.15	⟶	92.22 ± 0.10	2.06 / 21%
SST-5	McCann et al. (2017)	53.7	51.4	⟶	54.7 ± 0.5	3.3 / 6.8%

accuracy

F_1

average F_1

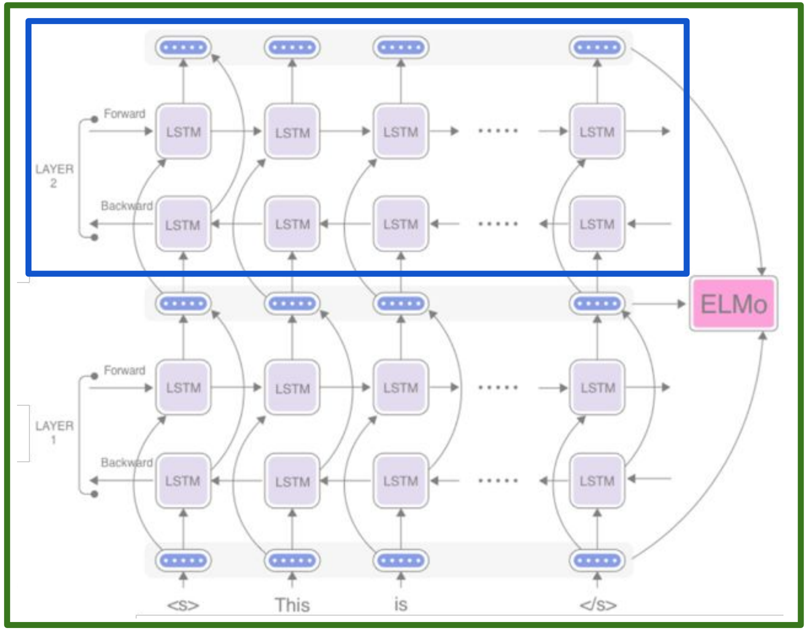
SOTA: state-of-the-art

5 Analysis

- 5.1. Alternate layer weighting schemes

Task	Baseline	Last Only	All layers λ=1	All layers λ=0.001
SQuAD	80.8	84.7	85.0	**85.2**
SNLI	88.1	89.1	89.3	**89.5**
SRL	81.6	84.1	84.6	**84.8**

Table 2: Development set performance for SQuAD, SNLI and SRL comparing using all layers of the biLM (with different choices of regularization strength λ) to just the top layer.



5 Analysis

- 5.2. Where to include ELMo?

Task	Input Only	Input & Output	Output Only
SQuAD	85.1	**85.6**	84.8
SNLI	88.9	**89.5**	88.7
SRL	**84.7**	84.3	80.9

Table 3: Development set performance for SQuAD, SNLI and SRL when including ELMo at different locations in the supervised model.

5 Analysis

- 5.3. What information is captured by the biLM's representations?

	Source	Nearest Neighbors
GloVe	play	playing, game, games, played, players, plays, player, Play, football, multiplayer
biLM	Chico Ruiz made a spectacular play on Alusik 's grounder {...}	Kieffer , the only junior in the group , was commended for his ability to hit in the clutch , as well as his all-round excellent play .
	Olivia De Havilland signed to do a Broadway play for Garson {...}	{...} they were actors who had been handed fat roles in a successful play , and had talent enough to fill the roles competently , with nice understatement .

Table 4: Nearest neighbors to "play" using GloVe and the context embeddings from a biLM.

5 Analysis

- 5.3. What information is captured by the biLM's representations?

Model	F_1
WordNet 1st Sense Baseline	65.9
Raganato et al. (2017a)	69.9
Iacobacci et al. (2016)	**70.1**
CoVe, First Layer	59.4
CoVe, Second Layer	*64.7*
biLM, First layer	67.4
biLM, Second layer	*69.0*

Table 5: All-words fine grained WSD F_1. For CoVe and the biLM, we report scores for both the first and second layer biLSTMs.

Model	Acc.
Collobert et al. (2011)	97.3
Ma and Hovy (2016)	97.6
Ling et al. (2015)	**97.8**
CoVe, First Layer	*93.3*
CoVe, Second Layer	92.8
biLM, First Layer	*97.3*
biLM, Second Layer	96.8

Table 6: Test set POS tagging accuracies for PTB. For CoVe and the biLM, we report scores for both the first and second layer biLSTMs.

5 Analysis

- 5.4. Sample efficiency

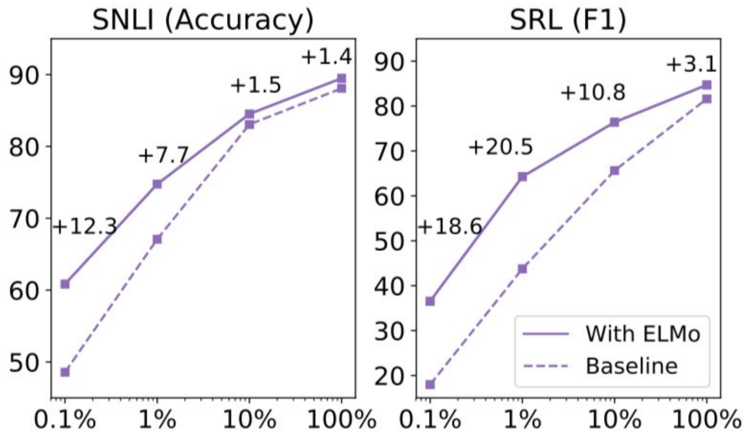


Figure 1: Comparison of baseline vs. ELMo performance for SNLI and SRL as the training set size is varied from 0.1% to 100%.

- 5.5. Visualization of learned weights

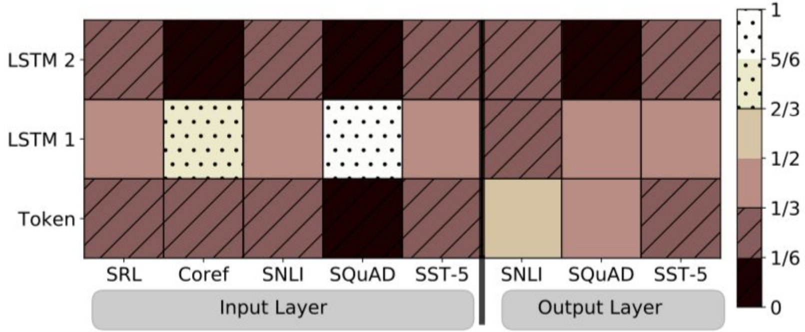


Figure 2: Visualization of softmax normalized biLM layer weights across tasks and ELMo locations. Normalized weights less then 1/3 are hatched with horizontal lines and those greater then 2/3 are speckled.

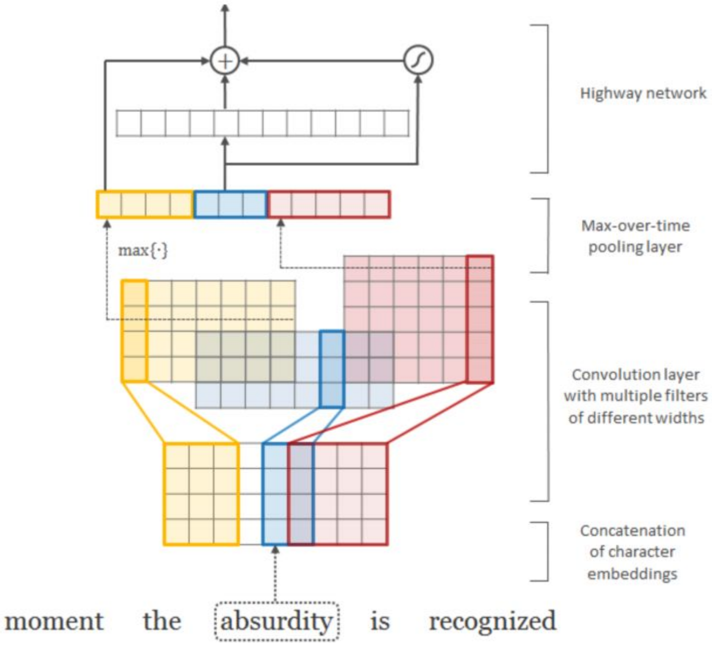
6 Conclusion

- We have introduced a general approach for learning **high-quality deep context-dependent representations from biLMs**, and shown large improvements when applying ELMo to a broad range of NLP tasks.

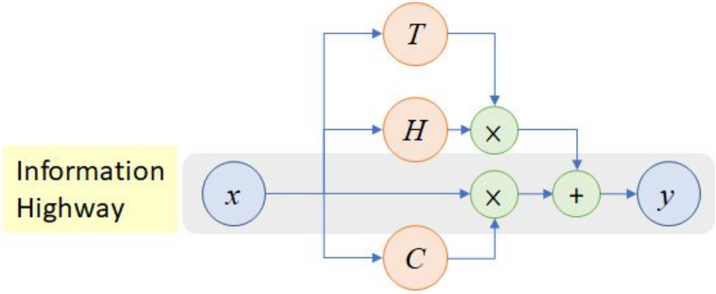
- Through ablations and other controlled experiments, we have also confirmed that the **biLM layers efficiently encode different types of syntactic and semantic information** about words in-context, and that using all layers improves overall task performance.

Appendix

3 ELMo Input



Character-aware neural language models



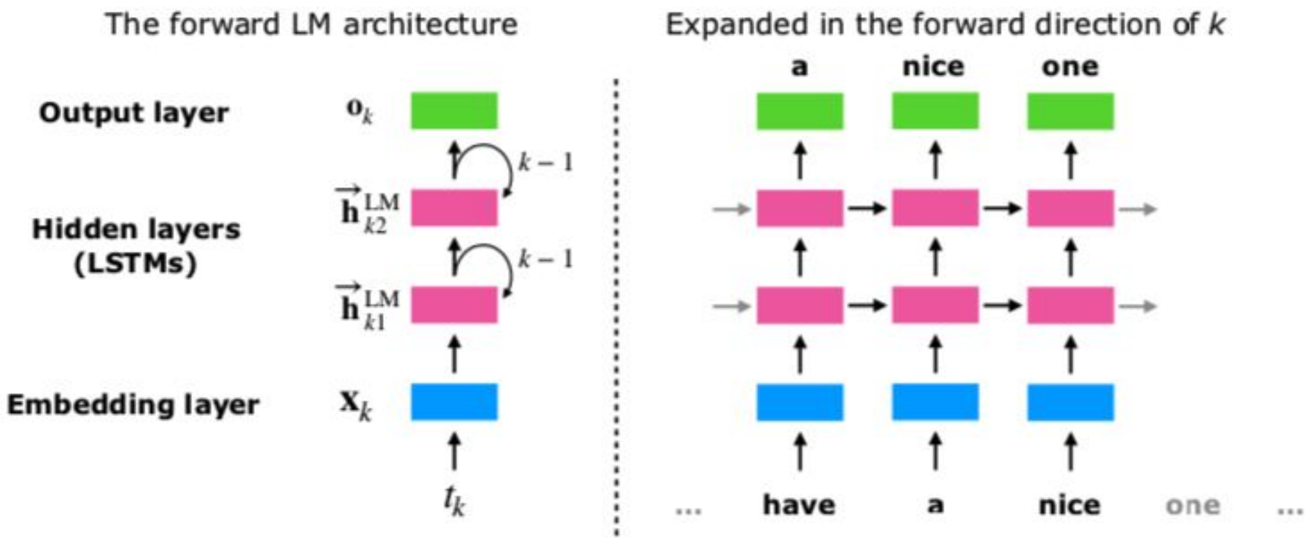
H: activation / T: transform gate / C: carry gate (= 1-T)

- Highway network은 LSTM의 구조에서 착안하여 각 층들에 대하여 gate를 학습하는 장치를 추가
- x, y, H(x, W), T(x, W)의 dimension이 일치해야 함
  → 1) x를 subsampling하여 크기 감소
  → 2) zero-padding하여 크기가 감소하지 않도록 함
- MNIST와 CIFAR100 데이터에 대하여 기존 네트워크는 깊이가 깊어질 수록 오히려 성능이 떨어지는 경우가 있는 반면 Highway network은 더 좋은 성능을 낼 수 있었다

http://mlgalaxy.blogspot.com/2018/12/elmo-deep-contextualized-word.html
https://lyusungwon.github.io/deep-learning/2018/06/05/hn.html

3 ELMo BiLSTM

With long short term memory (LSTM) network, predicting the next words in both directions to build biLMs



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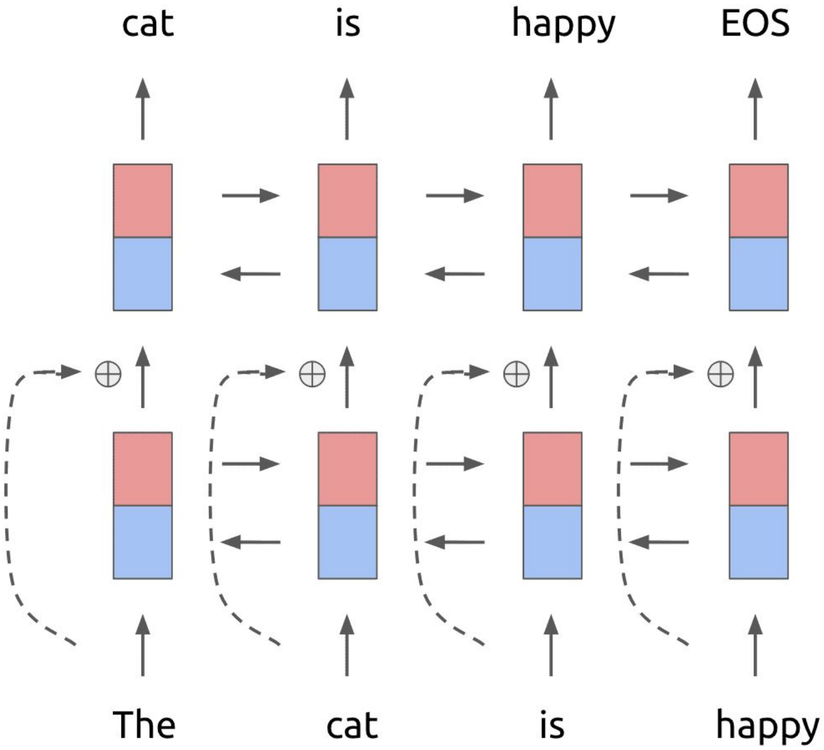
3 ELMo biLSTM



https://www.mihaileric.com/posts/deep-contextualized-word-representations-elmo/

4 Evaluation

- Baseline
  - SQuAD - Improved version of the Bidirectional Attention Flow model ()
  - SNLI: ESIM sequence model
  - SRL - He et al. (2017) modelded SRL as a BIO tagging problem and used an 8-layer deep biLSTM with forward and backward directions interleaved, flollowing Zhou and Xu(2015).
  - Coref - End-to=end span-based neural model
  - NER - Baseline model used pre-trained word embeddings, a character-based CNN representation, two biLSTM layers and a conditional random field (CRF) loss, similar to Colobert et al.
  - SST-5 - Biattentive classification network